

SARA ARIF

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PROFESSIONAL PROFILE

Production-oriented Computer Science undergraduate (CGPA 3.85/4.0, ranked 2nd in CS Department, University of Chakwal) with deep hands-on engineering experience across full-stack web and mobile development, machine learning pipelines, ETL data engineering, and multi-agent AI system orchestration. Designs and ships systems used in real environments — from an offline-first PWA serving 50+ construction workers to LALR(1) compiler simulators, LangGraph agent backends, and custom language toolchains. Committed to engineering that reaches production, not just Jupyter notebooks.

Silver Honour recipient — Top 2% globally at ICSC 2025 among 3,800+ international participants. PM Laptop Scheme 2025 awardee — ranked 2nd in CS Department and 27th university-wide out of 6,000+ students.

EDUCATION

University of Chakwal — Bachelor of Science in Computer Science Sep 2023 – Sep 2027 (Expected)
CGPA: 3.85 / 4.0 | Ranked 2nd in CS Department | PM Laptop Scheme 2025 — 27th university-wide out of 6,000+ students

Qadeer Girls College — Intermediate, Mathematics & Computer Science May 2021 – Apr 2023
Grade: A+ | Cricket Team Captain, Orator

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, SQL (MySQL, PostgreSQL), C++, C#, PHP

AI / Machine Learning: Scikit-learn, CatBoost, TensorFlow, SHAP, SMOTE, LangGraph, RAG, HuggingFace, Groq, Streamlit

Data Engineering & BI: Pandas, NumPy, ETL Pipeline Design, Power BI, Advanced DAX, Tableau, Snowflake, Microsoft Fabric, SQLite

Web & Mobile Development: React, React Native, FastAPI, REST API Design, Progressive Web Apps (PWA), PHP/MySQL, Service Workers

Compiler & Systems: Lexer/Parser Design, AST Construction, LALR(1) Parsing, Intermediate Representation, Code Generation

Developer Tools: Git, Docker, Vercel, Figma, Cisco Packet Tracer, Tkinter, SQLite

TECHNICAL PROJECTS

VizBot Analytics (DataSage) · LangGraph · Groq Llama 3.1 · FastAPI · Python

Problem Statement

Data analysts spend significant time manually profiling datasets, identifying anomalies, and writing narrative summaries. No unified natural-language interface existed to automate end-to-end exploratory data analysis for arbitrary uploaded datasets.

Architecture & Implementation

- Orchestrated a multi-agent LangGraph pipeline with discrete agents for: data profiling, statistical anomaly detection, visualization generation, and natural-language narrative synthesis — triggered by a single user query.
- Decoupled LLM inference (Groq Llama 3.1) into an async FastAPI backend, achieving non-blocking concurrent request execution and eliminating UI-side latency under load.
- Implemented sandboxed code execution for all agent-generated Python visualization scripts, preventing arbitrary code from accessing the host environment.

Outcomes

- Reduced analyst setup time for structured EDA from hours to minutes; pipeline handles profiling, anomaly detection, and narrative generation with zero manual intervention between stages.

AURA Chat Model · LangGraph · HuggingFace 120B · SQLite · Python

Problem Statement

Standard chatbot implementations lack persistent cross-session memory and thread isolation, causing context loss between sessions and cross-user state leakage in multi-user deployments.

Architecture & Implementation

- Built stateful conversational AI using LangGraph StateGraph with a 120B-parameter HuggingFace model, delivering streaming token-level responses for responsive UX.

- Engineered cross-session memory persistence via SQLite checkpointing with complete thread isolation — each user session maintains independent state with no leakage across threads.

Outcomes

- Delivered production-grade session persistence and isolation; eliminated context-loss failures common in stateless chatbot deployments.

Bank Loan Risk Classifier · Python · CatBoost · SHAP · SMOTE · Streamlit

Problem Statement

Loan default classification datasets exhibit severe class imbalance (as low as 10% positive class), causing standard classifiers to optimize for accuracy at the cost of recall — missing true default cases and exposing lenders to unquantified risk.

Architecture & Implementation

- Applied SMOTE oversampling to correct a 90/10 class imbalance; trained CatBoost classifier with Bayesian hyperparameter tuning optimizing for recall on the minority (default) class.
- Integrated SHAP for per-prediction interpretability, generating waterfall plots that show each feature's contribution to individual predictions for non-technical audit.
- Deployed as an interactive Streamlit dashboard with real-time prediction interface and embedded SHAP visualization per submitted loan application.

Outcomes

- Achieved 30%+ reduction in false negatives versus baseline classifier — directly measurable improvement in risk-adjusted lending decision quality.
- Reduced stakeholder model review cycles by 50% through SHAP-powered explainability, enabling non-technical compliance teams to audit model outputs independently.

Customer Trends ETL Pipeline · Python · SQL · Pandas · Power BI · Advanced DAX

Problem Statement

Business analytics teams relied on a manual weekly reporting process — SQL extraction, spreadsheet manipulation, and manual chart construction — consuming significant analyst time and introducing human error in aggregation.

Architecture & Implementation

- Designed a full 3-layer ETL pipeline: SQL extraction from relational source → Python transformation layer (feature engineering, null handling, normalization, outlier treatment) → automated load into Power BI data model.
- Implemented Advanced DAX measures for time intelligence, YoY comparisons, and rolling averages across the executive dashboard.
- Modularized Python transformation code into reusable, parameterized functions, enabling rapid onboarding of new data sources without pipeline refactoring.

Outcomes

- Compressed weekly reporting turnaround from a multi-day manual process to real-time automated refresh, eliminating analyst involvement in routine aggregation tasks.

UrduPy Compiler · Python · Tkinter · Compiler Theory

Problem Statement

230M+ Urdu speakers have no programming tooling in their native language. Existing CS education resources are exclusively English, creating a systemic accessibility barrier for non-English speakers learning to code.

Architecture & Implementation

- Designed and implemented a complete 5-phase compiler from scratch: Lexer (tokenization) → Recursive Descent Parser → Abstract Syntax Tree (AST) construction → Intermediate Representation (IR) generation → target Code Generation.
- Implemented semantic analysis and type-checking passes, detecting type mismatches and undeclared variable errors at compile time rather than runtime.
- Packaged with a custom Tkinter GUI IDE featuring real-time type inference, syntax highlighting, bracket matching, and inline error reporting at the token level.

Outcomes

- Delivered the first functional programming toolchain designed specifically for Roman Urdu, providing an accessible path to programming for native Urdu speakers without English prerequisite.

SQL2Pandas Transpiler · Python · AST Parsing · Compiler Optimization

Problem Statement

Data analysts fluent in SQL face a productivity barrier when moving to Python workflows — manual translation of SQL queries to Pandas chains is error-prone, time-consuming, and produces inefficient, non-idiomatic code.

Architecture & Implementation

- Implemented a 6-phase transpiler: Lexer → Parser → AST → Semantic Analysis → Optimizer → Code Generator, with full AST support for 20+ SQL constructs: JOINS, subqueries, GROUP BY with HAVING, window functions, ORDER BY with LIMIT.

- Applied 5 compiler optimizer passes — predicate pushdown, projection pruning, constant folding, dead code elimination, and loop-invariant hoisting — to reduce computation in generated Pandas output.

Outcomes

- Produced idiomatic, optimized Pandas code from raw SQL queries, reducing manual translation effort to zero and improving generated code efficiency through optimizer passes.

LALR(1) Compiler Simulator · *React · JavaScript · Compiler Theory — lalr-simulator.vercel.app*

Problem Statement

LALR(1) parsing — the algorithm underlying most production language parsers — is notoriously difficult to understand from static textbook diagrams. Students lack interactive tools to observe item set construction and parse table computation in real time.

Architecture & Implementation

- Implemented the complete LALR(1) algorithm from scratch: canonical LR(0) item set construction with closure and GOTO operations, lookahead propagation, and shift-reduce/reduce-reduce conflict detection and resolution.
- Built a live React visualizer (deployed at lalr-simulator.vercel.app) with step-by-step parse table construction, animated parse stack, and input buffer rendering for interactive exploration.

Outcomes

- Deployed live at lalr-simulator.vercel.app; provides an interactive learning environment for compiler theory students to observe and verify LALR(1) algorithm execution on arbitrary grammars.

DG Cement Work Tracker · *PHP · MySQL · JavaScript · PWA · REST APIs*

Problem Statement

A cement construction firm managing 50+ field workers across remote sites relied entirely on paper-based attendance sheets and manual payroll calculation, resulting in data loss, calculation errors, and multi-day payroll processing delays.

Architecture & Implementation

- Architected a full-stack Progressive Web App (PWA) with 28 REST API endpoints (PHP/MySQL) serving a mobile-first supervisor interface for real-time attendance and payroll logging.
- Implemented offline-first service worker caching strategy, ensuring zero data loss at construction sites with no internet connectivity — data syncs automatically on reconnection.
- Designed a normalized MySQL schema with relational integrity constraints supporting multi-supervisor, multi-site deployment with audit-ready payroll records.

Outcomes

- Replaced 100% of paper-based records for 50+ workers — currently live and in active production use at the construction site.
- Eliminated payroll calculation errors and reduced administrative processing time by an estimated 70%+ per weekly payroll cycle.

Electricity Consumption Forecasting Dashboard · *Python · Scikit-learn · Streamlit · Time-Series ML*

Problem Statement

Utility stakeholders lack accessible tools for monitoring predicted electricity demand against actual consumption, making it difficult to identify seasonal anomalies or demand spikes in time for resource reallocation.

Architecture & Implementation

- Engineered a time-series ML forecasting pipeline with temporal feature engineering: lag variables, rolling window statistics, hour-of-day / day-of-week encodings, and seasonal decomposition.
- Deployed as a live Streamlit dashboard with dynamic predicted vs. actual consumption charts, configurable forecast horizons, and anomaly flagging for demand spikes.

Outcomes

- Delivered an end-to-end forecasting system enabling stakeholders to self-serve consumption predictions and identify demand anomalies without requiring data science expertise.

Enterprise Networking Simulation · *Cisco Packet Tracer · OSPF · VLAN · Network Security*

Problem Statement

Designing a hierarchical, secure enterprise network for a 240-employee banking firm requires rigorous VLAN segmentation, dynamic routing, and access control policies — typically requiring physical hardware for testing.

Architecture & Implementation

- Simulated a 3-tier hierarchical network (core, distribution, access layers) using Cisco Packet Tracer, with OSPF dynamic routing configured across all inter-VLAN boundaries.
- Implemented VLAN segmentation for departmental isolation (HR, Finance, Operations, IT), access control lists (ACLs) for inter-VLAN traffic policy, and port security for endpoint protection.

Outcomes

- Produced a fully functional network simulation validating routing convergence, VLAN isolation integrity, and security policy enforcement for a 240-employee banking infrastructure.

WORK EXPERIENCE

Freelance Software Architect — Self-Employed (Remote Contract)

May 2026 – Present

- Engaged directly by DG Cement firm to architect and deploy a full production payroll and attendance management system — scoped, designed, and shipped end-to-end as sole developer.
- Architected offline-first Progressive Web App serving 50+ field workers across remote construction sites: 28 REST API endpoints (PHP/MySQL), service worker caching for zero data loss during internet outages, mobile-first supervisor interface.
- Replaced 100% of the firm's paper-based tracking with a live digital system, eliminating calculation errors and compressing weekly payroll processing time by an estimated 70%+.
- Designed normalized relational database schema supporting multi-site deployment, audit trails, and historical payroll record retrieval.

Machine Learning Intern — DevelopersHub Corporation (Remote, Full-Time)

Jun 2025 – Jul 2025

- Conducted production-grade EDA on raw business datasets — systematic detection of outliers, null patterns, data type inconsistencies, and seasonal distribution shifts to produce clean, model-ready feature matrices.
- Resolved a critical 90/10 class imbalance using SMOTE oversampling; designed and tuned a CatBoost gradient boosting classifier with Bayesian hyperparameter search to optimize recall on the minority (default) class.
- Achieved 30%+ reduction in false negatives versus baseline — directly improving the risk sensitivity of business-critical loan approval decisions.
- Implemented SHAP (SHapley Additive exPlanations) interpretability framework; generated per-prediction waterfall plots and aggregate feature importance reports that reduced stakeholder model review cycles by 50%.

PROFESSIONAL CERTIFICATIONS

- Generative AI Professional — Snowflake
- Advanced Deep Learning Specialist — IBM / Coursera
- AI Agents Using RAG and LangChain — IBM / Coursera
- ChatGPT Advanced Data Analysis — Vanderbilt University / Coursera
- Getting Started with Microsoft Fabric — Whizlabs / Coursera
- Generative AI: Elevate Your Data Science Career — IBM / Coursera
- Building Voice Agents with Vapi — DataCrumbs
- Vibe Coding with Cursor AI — United Latino Students Association

RELEVANT COURSEWORK

Data Structures & Algorithms | Compiler Construction | Artificial Intelligence | Operating Systems | Database Systems | Computer Networks | Statistical Methods | Software Engineering

ACHIEVEMENTS & AWARDS

- Silver Honour — Top 2% globally at ICSC 2025, ranked among 3,800+ participants from institutions worldwide.
- PM Laptop Scheme 2025 — Ranked 2nd in CS Department and 27th university-wide out of 6,000+ students; merit-awarded by the Government of Pakistan.

RESEARCH & PROFESSIONAL INTERESTS

Multi-agent AI system orchestration and LLM backends | Production ETL pipeline engineering and data automation | Compiler theory, language design, and formal methods | Cross-platform mobile development | Open-source AI tooling and developer experience

PORTFOLIO & LINKS

Portfolio: sara-3d-portfolio.vercel.app GitHub: github.com/SaraArif6198 LinkedIn: linkedin.com/in/sara-arif-792p LALR Simulator: lalr-simulator.vercel.app